12TH CHEMISTRY · CHAPTER 31 · FULL CHAPTER MCQS

Materials and Energy

123 topic-wise MCQs from the complete chapter with answers and reasons

31.1 MATERIALS: METALS AND ALLOYS

1. The energy device in a mobile phone is a:

(a) fuel cell (b) lead-acid battery
(c) dry cell (d) Li-ion battery

Answer: (d) Li-ion battery — Figure 31.1.

2. Metals can be drawn into wires because they are:

(a) ductile (b) brittle
(c) volatile (d) insulating

Answer: (a) ductile — Stated in the book.

3. Aluminium is used in aircraft bodies because it is:

(a) the hardest metal (b) the best conductor
(c) light-weight (d) magnetic

Answer: (c) light-weight — Stated in the book.

4. Copper is used in electric wires because of its:

(a) low density (b) excellent conductivity
(c) high melting point (d) colour

Answer: (b) excellent conductivity — Stated in the book.

5. Steel is used in bridges because of its:

(a) transparency (b) low cost only
(c) softness (d) strength

Answer: (d) strength — Stated in the book.

6. A metal found in the free state is:

(a) gold (b) sodium
(c) aluminium (d) iron

Answer: (a) gold — It is less reactive.

7. Haematite is an ore of:

(a) copper (b) iron
(c) zinc (d) aluminium

Answer: (b) iron — Fe_2O_3.

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8. The earthy impurities present in an ore are called:

(a) alloy (b) slag
(c) gangue (d) cullet

Answer: (c) gangue — Stated in the book.

9. The science of extracting metals from their ores is:

(a) metallurgy (b) crystallography
(c) spectroscopy (d) petrology

Answer: (a) metallurgy — Definition.

10. Freeing the ore from impurities is called:

(a) roasting (b) ore concentration
(c) refining (d) alloying

Answer: (b) ore concentration — Step ii.

11. Roasting is the heating of a:

(a) molten salt with current (b) carbonate ore without air
(c) metal oxide with carbon (d) sulphide ore in excess air

Answer: (d) sulphide ore in excess air — Definition.

12. Calcination is the heating of a:

(a) pure metal in oxygen (b) sulphide ore in excess air
(c) carbonate ore in limited or no air (d) metal with acid

Answer: (c) carbonate ore in limited or no air — Definition.

13. The gas released when zinc sulphide is roasted is:

(a) SO_2 (b) CO_2
(c) H_2 (d) O_2

Answer: (a) SO_2 — $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$.

14. The gas released when zinc carbonate is calcined is:

(a) SO_2 (b) CO_2
(c) CO (d) H_2

Answer: (b) CO_2 — $ZnCO_3 \rightarrow ZnO + CO_2$.

15. The actual reducing agent for iron oxide in the furnace is:

(a) hydrogen (b) carbon dioxide
(c) oxygen (d) carbon monoxide

Answer: (d) carbon monoxide — $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$.

16. Metals more reactive than carbon are extracted by:

(a) roasting only (b) reduction with carbon
(c) electrolysis (d) calcination only

Answer: (c) electrolysis — Carbon cannot reduce their oxides.

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17. Which metal is extracted by electrolysis?

(a) aluminium (b) iron
(c) zinc (d) gold

Answer: (a) aluminium — It is more reactive than carbon.

18. In electrolysis, metal ions are discharged at the:

(a) electrolyte surface (b) anode
(c) salt bridge (d) cathode

Answer: (d) cathode — Positive ions gain electrons there.

19. Copper for electrical wiring must be:

(a) 90% pure (b) 99.99% pure
(c) 98% pure (d) 50% pure

Answer: (b) 99.99% pure — Stated in the book.

20. Brass is an alloy of:

(a) iron and carbon (b) copper and tin
(c) copper and zinc (d) aluminium and copper

Answer: (c) copper and zinc — Table 31.1.

21. Bronze is an alloy of:

(a) copper and zinc (b) copper and tin
(c) iron and nickel (d) aluminium and magnesium

Answer: (b) copper and tin — Table 31.1.

22. Bronze is used for:

(a) medals and statues (b) aircraft bodies
(c) electric wires (d) spark plugs

Answer: (a) medals and statues — Hard and wear-resistant.

23. The main non-metal in steel is:

(a) phosphorus (b) sulphur
(c) silicon (d) carbon

Answer: (d) carbon — 0.1–2%.

24. The alloy used for aircraft bodies is:

(a) bronze (b) brass
(c) duralumin (d) steel

Answer: (c) duralumin — Lightweight and strong.

25. The chief metal in duralumin is:

(a) manganese (b) copper
(c) magnesium (d) aluminium

Answer: (d) aluminium — 93–95%.

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26. Alloying changes the _____ of a metal.

(a) microstructure (b) atomic number
(c) isotopes (d) nuclear charge

Answer: (a) microstructure — To enhance its properties.

31.1 CERAMICS, COMPOSITES, SEMICONDUCTORS

27. Ceramics are:

(a) malleable (b) soft and ductile
(c) hard but brittle (d) good conductors

Answer: (c) hard but brittle — Stated in the book.

28. Ceramics are suitable for furnaces because of their:

(a) electrical conductivity (b) high melting temperatures
(c) flexibility (d) low density

Answer: (b) high melting temperatures — Heat resistance.

29. Spark plugs are made from:

(a) silicon carbide (b) porcelain only
(c) glass (d) alumina

Answer: (d) alumina — Stated in the book.

30. The abrasive used in sandpaper is:

(a) silicon carbide (b) alumina
(c) porcelain (d) silica gel

Answer: (a) silicon carbide — Table 31.2.

31. Insulators on power lines are made of:

(a) fibreglass (b) copper
(c) porcelain (d) silicon

Answer: (c) porcelain — Table 31.2.

32. Optical fibres are made of:

(a) steel (b) glass
(c) bronze (d) rubber

Answer: (b) glass — Stated in the book.

33. A composite consists of a matrix and:

(a) monomers (b) catalysts
(c) solvents (d) reinforcement agents

Answer: (d) reinforcement agents — Definition.

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34. In a composite, strength is provided mainly by the:

(a) reinforcement (b) matrix
(c) solvent (d) pigment

Answer: (a) reinforcement — The matrix holds it.

35. The matrix in fibreglass is a:

(a) ceramic (b) metal
(c) polymer (d) semiconductor

Answer: (c) polymer — Stated in the book.

36. Boat hulls and car bodies are made of:

(a) bronze (b) fibreglass
(c) porcelain (d) silicon

Answer: (b) fibreglass — Stated in the book.

37. Carbon fibre is widely used in:

(a) fertilizers (b) power line insulators
(c) food packaging (d) aircraft and sports gear

Answer: (d) aircraft and sports gear — Light and durable.

38. Composites have a good:

(a) strength-to-weight ratio (b) electrical conductivity
(c) solubility in water (d) melting behaviour

Answer: (a) strength-to-weight ratio — Stated in the book.

39. The most common semiconductor is:

(a) germanium oxide (b) gallium arsenide
(c) copper (d) silicon

Answer: (d) silicon — Used in microchips and solar panels.

40. A mobile SIM card is mainly composed of a chip of:

(a) silicon (b) copper
(c) gold (d) alumina

Answer: (a) silicon — With copper connectors.

41. Gallium arsenide is used in:

(a) boat hulls (b) furnace bricks
(c) 5G wireless communication devices (d) medals

Answer: (c) 5G wireless communication devices — Stated in the book.

42. Gallium nitride is used in electric cars for:

(a) tyres (b) fast charging
(c) body panels (d) windows

Answer: (b) fast charging — Advanced conductivity.

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43. In a semiconductor, the current:

(a) flows without any control (b) never flows
(c) can be controlled as required (d) flows only when molten

Answer: (c) can be controlled as required — The difference from conductors.

31.2 TOXICITY AND RECYCLING

44. The extent to which a substance can harm an organism is its:

(a) solubility (b) density
(c) reactivity (d) toxicity

Answer: (d) toxicity — Definition.

45. A substance that causes cancer is a:

(a) catalyst (b) mutagen
(c) carcinogen (d) pathogen

Answer: (c) carcinogen — Stated in the book.

46. A substance that changes DNA and genes is a:

(a) carcinogen (b) mutagen
(c) allergen (d) antigen

Answer: (b) mutagen — Stated in the book.

47. A metal described as a carcinogen and mutagen is:

(a) chromium (b) sodium
(c) calcium (d) magnesium

Answer: (a) chromium — Health hazard example.

48. Flammable and explosive materials pose:

(a) nutritional hazards (b) no hazard
(c) physical hazards (d) optical hazards

Answer: (c) physical hazards — Stated in the book.

49. Which is NOT a heavy metal named in the book?

(a) lead (b) sodium
(c) mercury (d) cadmium

Answer: (b) sodium — Pb, Hg and Cd are named.

50. Heavy metals are commonly found in:

(a) electronics and batteries (b) fruits
(c) rain water (d) wood

Answer: (a) electronics and batteries — And fluorescent lights.

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51. BPA stands for:

(a) bromo propanoic acid (b) benzene phenyl acid
(c) butyl propyl amine (d) bisphenol A

Answer: (d) bisphenol A — A plastic additive.

52. Phthalates and BPA interfere with:

(a) bones (b) hormones
(c) eyesight (d) hearing

Answer: (b) hormones — Stated in the book.

53. Tiny plastic particles found in oceans, soil and food are:

(a) cullet (b) monomers
(c) microplastics (d) gangue

Answer: (c) microplastics — Stated in the book.

54. Recycling aluminium saves about _____ of the energy.

(a) 50% (b) 5%
(c) 25% (d) 95%

Answer: (d) 95% — Compared with extraction from ore.

55. Mixed materials in phones and laptops cause the challenge of:

(a) material separation (b) over-supply
(c) low demand (d) high purity

Answer: (a) material separation — First challenge.

56. Contaminated recycled plastic cannot be used for:

(a) pipes (b) road material
(c) food-grade products (d) benches

Answer: (c) food-grade products — Safety reasons.

57. Breaking polymers into monomers is:

(a) chemical recycling (b) mechanical recycling
(c) pulping (d) calcination

Answer: (a) chemical recycling — Depolymerization.

58. Mechanical recycling of plastics is limited by:

(a) lack of plastics
(b) contamination and chain degradation
(c) low melting point of glass
(d) the use of catalysts

Answer: (b) contamination and chain degradation — Table 31.3.

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59. Paper is recycled by:

(a) electrolysis
(b) cullet melting
(c) depolymerization
(d) pulping and de-inking

Answer: (d) pulping and de-inking — Table 31.3.

60. Inks are removed from paper pulp using:

(a) HCl or H_2SO_4
(b) NaOH or Na_2CO_3
(c) NaCl
(d) $CaCO_3$

Answer: (b) NaOH or Na_2CO_3 — Table 31.3.

61. Scrap glass prepared for recycling is called:

(a) cullet
(b) gangue
(c) slag
(d) pulp

Answer: (a) cullet — Table 31.3.

62. Glass is a:

(a) polymer of ethene
(b) crystalline metal
(c) non-crystalline solid
(d) liquid metal

Answer: (c) non-crystalline solid — So its recycling is chemically simpler.

63. Recycling reduces the solid waste going to:

(a) power plants
(b) oceans only
(c) the atmosphere
(d) landfills

Answer: (d) landfills — Stated in the book.

31.3 X-RAY CRYSTALLOGRAPHY

64. X-ray crystallography gives the:

(a) mass of an isotope
(b) boiling point of a liquid
(c) colour of a gas
(d) three-dimensional positions of atoms in a crystal

Answer: (d) three-dimensional positions of atoms in a crystal — Definition.

65. X-rays are suitable for locating atoms because of their:

(a) very long wavelength
(b) very small wavelength
(c) low energy
(d) visible colour

Answer: (b) very small wavelength — Stated in the book.

66. When X-rays hit the atoms of a crystal they are:

(a) converted to sound (b) absorbed completely
(c) diffracted (d) stopped

Answer: (c) diffracted — At specific angles.

67. The pattern formed on the screen is called a:

(a) diffraction pattern (b) chromatogram
(c) mass spectrum (d) thermogram

Answer: (a) diffraction pattern — Stated in the book.

68. The path length difference in X-ray diffraction is:

(a) 2d cosθ (b) d sinθ
(c) 2d sinθ (d) d/2 sinθ

Answer: (c) 2d sinθ — Figure 31.4.

69. The best-known use of XRD was to determine the structure of:

(a) DNA (b) glucose
(c) water (d) methane

Answer: (a) DNA — The double helix.

70. In drug development XRD shows how a drug:

(a) changes colour (b) tastes
(c) dissolves in water (d) attaches to its target

Answer: (d) attaches to its target — Stated in the book.

71. The molecule whose diffraction pattern is shown in the book is:

(a) benzene (b) naphthalene
(c) glucose (d) ethanol

Answer: (b) naphthalene — Figure 31.5.

72. XRD requires the sample to be a:

(a) crystalline solid (b) gas
(c) liquid (d) solution only

Answer: (a) crystalline solid — Regular layers of atoms are needed.

31.4–31.5 FOSSIL FUELS AND DISTILLATION

73. Which is NOT a fossil fuel?

(a) uranium (b) coal
(c) petroleum (d) natural gas

Answer: (a) uranium — It is a nuclear fuel.

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74. The major component of natural gas is:

(a) propane (b) ethane
(c) methane (d) butane

Answer: (c) methane — CH_4.

75. Methane is mainly used to produce:

(a) glass and paper (b) benzene and toluene
(c) steel and cement (d) ammonia and methanol

Answer: (d) ammonia and methanol — Stated in the book.

76. Coal gives which chemicals for dyes and explosives?

(a) methane and ethane (b) benzene, toluene and xylene
(c) ethene and propene (d) ammonia and methanol

Answer: (b) benzene, toluene and xylene — Stated in the book.

77. Coal is mainly used for:

(a) electricity generation, cement and steel
(b) cooking gas
(c) jet fuel
(d) lubricants

Answer: (a) electricity generation, cement and steel — Table 31.4.

78. Burning fossil fuels releases the greenhouse gas:

(a) nitrogen (b) oxygen
(c) carbon dioxide (d) argon

Answer: (c) carbon dioxide — Leading to global warming.

79. Crude oil is separated by:

(a) chromatography (b) filtration
(c) electrolysis (d) fractional distillation

Answer: (d) fractional distillation — A physical method.

80. Fractional distillation depends on differences in:

(a) colours (b) boiling points
(c) densities only (d) solubilities in water

Answer: (b) boiling points — Stated in the book.

81. In a fractionating column the temperature:

(a) is highest at the top (b) increases going up
(c) is the same everywhere (d) decreases going up

Answer: (d) decreases going up — Stated in the book.

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82. Heavier hydrocarbons condense:

(a) near the bottom of the column (b) at the top
(c) outside the column (d) in the furnace

Answer: (a) near the bottom of the column — They have higher boiling points.

83. The fraction collected at the top of the column is:

(a) bitumen (b) diesel
(c) gases (LPG) (d) lubricating oil

Answer: (c) gases (LPG) — Lowest boiling.

84. The fraction used as jet fuel is:

(a) gasoline (b) kerosene
(c) diesel (d) bitumen

Answer: (b) kerosene — Figure 31.6.

85. The residue used for road surfacing is:

(a) bitumen (b) kerosene
(c) naphtha (d) LPG

Answer: (a) bitumen — Figure 31.6.

86. Diesel is used in:

(a) cooking stoves only (b) lamps
(c) trucks, buses and generators (d) road surfacing

Answer: (c) trucks, buses and generators — Figure 31.6.

31.6 PETROCHEMICALS

87. Petrochemicals are obtained mainly from:

(a) sea water (b) crude oil and natural gas
(c) air (d) limestone

Answer: (b) crude oil and natural gas — Definition.

88. The polymer obtained from ethene is used for:

(a) bags, bottles and packaging (b) explosives
(c) tyres (d) dyes

Answer: (a) bags, bottles and packaging — Poly(ethene).

89. TNT is made from:

(a) ethene (b) benzene
(c) xylene (d) toluene

Answer: (d) toluene — Trinitrotoluene.

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90. Polyester fibres are derived from:

(a) butadiene (b) toluene
(c) xylene (d) propene

Answer: (c) xylene — Table 31.5.

91. Synthetic rubber for tyres is made from:

(a) toluene (b) benzene
(c) ethene (d) butadiene

Answer: (d) butadiene — e.g. styrene butadiene rubber.

92. Polystyrene is derived from:

(a) xylene (b) propene
(c) benzene (via styrene) (d) butadiene

Answer: (c) benzene (via styrene) — Table 31.5.

93. Breaking large hydrocarbons into smaller molecules is:

(a) reforming (b) cracking
(c) roasting (d) alloying

Answer: (b) cracking — Definition.

94. Thermal cracking is carried out at:

(a) 450–750 °C and high pressure (b) 25 °C and low pressure
(c) 100 °C with a catalyst (d) 0 °C

Answer: (a) 450–750 °C and high pressure — Stated in the book.

95. The catalyst used in catalytic cracking is:

(a) iron (b) platinum
(c) nickel (d) zeolite

Answer: (d) zeolite — Aluminosilicates or silica-alumina.

96. A catalyst speeds up cracking by lowering the:

(a) temperature of the furnace (b) activation energy
(c) mass of the reactants (d) boiling point

Answer: (b) activation energy — Stated in the book.

97. Thermal cracking of decane gives octane and:

(a) ethene (b) propene
(c) methane (d) benzene

Answer: (a) ethene — $C_{10}H_{22} \rightarrow C_8H_{18} + C_2H_4$.

98. Thermal cracking normally gives a high percentage of:

(a) alcohols (b) alkanes only
(c) alkenes (d) aromatics

Answer: (c) alkenes — Important for plastics.

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99. Reforming is carried out mainly to:

(a) produce bitumen (b) remove sulphur
(c) make lubricants (d) raise the octane rating of fuel

Answer: (d) raise the octane rating of fuel — It avoids engine knocking.

100. Conversion of n-hexane into iso-hexane is:

(a) cracking (b) cyclization
(c) isomerisation (d) polymerization

Answer: (c) isomerisation — Branched isomers are formed.

101. Conversion of hexane into cyclohexane is:

(a) cyclization (b) isomerisation
(c) cracking (d) calcination

Answer: (a) cyclization — $C_6H_{14} \rightarrow C_6H_{12} + H_2$.

102. BTX stands for:

(a) butane, toluene and xenon (b) benzene, toluene and xylene
(c) butene, tin and xylene (d) benzene, tin and xenon

Answer: (b) benzene, toluene and xylene — Aromatic base chemicals.

103. The catalyst for forming benzene from hexane is:

(a) nickel (b) zeolite
(c) platinum (d) alumina

Answer: (c) platinum — Pt catalyst/heat.

104. A valuable by-product of reforming is:

(a) hydrogen (b) oxygen
(c) carbon dioxide (d) sulphur dioxide

Answer: (a) hydrogen — $C_6H_{14} \rightarrow C_6H_6 + 4H_2$.

31.7–31.8 ENERGY SOURCES

105. Energy density is the energy stored per unit:

(a) volume (b) mass
(c) time (d) area

Answer: (a) volume — MJ/L.

106. Specific energy is the energy contained per unit:

(a) time (b) volume
(c) mass (d) length

Answer: (c) mass — MJ/kg.

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107. The energy density of diesel is about:

(a) 3.5 MJ/L (b) 34.2 MJ/L
(c) 120 MJ/L (d) 35.8 MJ/L

Answer: (d) 35.8 MJ/L — Stated in the book.

108. The specific energy of hydrogen is:

(a) 35.8 MJ/kg (b) 120 MJ/kg
(c) 12 MJ/kg (d) 1.2 MJ/kg

Answer: (b) 120 MJ/kg — Very high.

109. Specific energy is most important in:

(a) home generators (b) power plants
(c) aviation and spacecraft (d) storage tanks

Answer: (c) aviation and spacecraft — Where weight is critical.

110. CNG is not ideal for long journeys because of its:

(a) lower energy density (b) high cost
(c) high specific energy (d) toxicity

Answer: (a) lower energy density — More storage volume is needed.

111. Splitting a heavy nucleus into lighter nuclei is:

(a) ionisation (b) fusion
(c) cracking (d) fission

Answer: (d) fission — Definition.

112. Combining light nuclei is:

(a) fission (b) fusion
(c) reforming (d) diffraction

Answer: (b) fusion — Definition.

113. A fuel used in nuclear fission is:

(a) uranium-235 (b) carbon-12
(c) hydrogen-1 (d) helium-4

Answer: (a) uranium-235 — Or plutonium-239.

114. Fission of uranium-235 gives barium-141, krypton-92 and:

(a) two electrons (b) one proton
(c) three neutrons (d) one alpha particle

Answer: (c) three neutrons — Plus energy.

115. The Chernobyl catastrophe occurred in:

(a) 2001 (b) 2011
(c) 1945 (d) 1986

Answer: (d) 1986 — Stated in the book.

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116. The Fukushima accident occurred in:

(a) 1986 (b) 2011
(c) 1979 (d) 2020

Answer: (b) 2011 — After an earthquake and tsunami.

117. A serious drawback of fission is:

(a) need for sunlight (b) greenhouse gases
(c) low energy output (d) radioactive waste

Answer: (d) radioactive waste — It must be handled for a long time.

118. Solar panels are typically made of:

(a) silicon (b) copper
(c) zeolite (d) alumina

Answer: (a) silicon — A semiconductor.

119. Solar panels convert sunlight to electricity by the:

(a) photoelectric emission of X-rays (b) greenhouse effect
(c) photovoltaic effect (d) thermal cracking

Answer: (c) photovoltaic effect — Stated in the book.

120. Solar panels produce:

(a) alternating current (AC) (b) direct current (DC)
(c) radio waves (d) heat only

Answer: (b) direct current (DC) — Converted later to AC.

121. DC is converted into AC by an:

(a) indicator (b) insulator
(c) injector (d) inverter

Answer: (d) inverter — Stated in the book.

122. Batteries in a solar system:

(a) convert DC to AC (b) produce sunlight
(c) store electricity for the night (d) cool the panels

Answer: (c) store electricity for the night — Also stabilise voltage.

123. Which is a renewable energy source?

(a) coal (b) solar energy
(c) petroleum (d) natural gas

Answer: (b) solar energy — Also hydro and wind.

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